1 High-Dimensional Classification: 1-NN Asymptotics

Consider two populations:

$$X_{11}, X_{12}, \ldots, X_{1n_1} \overset{iid}{\sim} N_d\left(\begin{pmatrix}0\\0\\\vdots\\0\end{pmatrix},\ \sigma_1^2 I\right)$$

$$X_{21}, X_{22}, \ldots, X_{2n_2} \overset{iid}{\sim} N_d\left(\begin{pmatrix}\mu\\\mu\\\vdots\\\mu\end{pmatrix},\ \sigma_2^2 I\right)$$

We have seen that:

(i) If $Z \sim N_d\left(\begin{pmatrix}0\\0\end{pmatrix}, \sigma_1^2 I\right)$, then

(a) $\frac{1}{d}\|Z - X_{1i}\|^2 \xrightarrow{p/as} 2\sigma_1^2$ for all $i = 1, 2, \ldots, n_1$

(b) $\frac{1}{d}\|Z - X_{2i}\|^2 \xrightarrow{p/as} \sigma_1^2 + \sigma_2^2 + \mu^2$ for all $i = 1, \ldots, n_2$

(ii) If $Z \sim N_d\left(\begin{pmatrix}\mu\\\vdots\\\mu\end{pmatrix}, \sigma_2^2 I\right)$, then

(a) $\frac{1}{d}\|Z - X_{1i}\|^2 \xrightarrow{p/as} \sigma_1^2 + \sigma_2^2 + \mu^2$ for $i = 1, 2, \ldots, n_1$

(b) $\frac{1}{d}\|Z - X_{2i}\|^2 \xrightarrow{p/as} 2\sigma_2^2$ for $i = 1, 2, \ldots, n_2$

Table 1: Limiting scaled squared distances

	$Z \sim N_d\big((0, \ldots, 0)^T, \sigma_1^2 I\big)$	$Z \sim N_d\big((\mu, \ldots, \mu)^T, \sigma_2^2 I\big)$
$\frac{1}{d}\|Z - X_{1i}\|^2$	$2\sigma_1^2$	$\sigma_1^2 + \sigma_2^2 + \mu^2$
$\frac{1}{d}\|Z - X_{2i}\|^2$	$\sigma_1^2 + \sigma_2^2 + \mu^2$	$2\sigma_2^2$

So, if $\mu^2 > |\sigma_1^2 - \sigma_2^2|$: The misclassification probability of the 1-NN classifier will converge to 0.
If $\mu^2 < |\sigma_1^2 - \sigma_2^2|$, it will classify all observations to one class.

In High Dimensions

If Z comes from $N_d\big((0, \ldots, 0)^T, \sigma_1^2 I\big)$ it will be correctly classified if

$$\mu^2 > \sigma_1^2 - \sigma_2^2$$

Similarly, if Z comes from $N\big((\mu, \ldots, \mu)^T, \sigma_2^2 I\big)$ it will be correctly classified if

$$\mu^2 > \sigma_2^2 - \sigma_1^2$$

2 Generalization (Hall, Marron, Neeman; 2005, JRSS-B)

$$X_{11}, X_{12}, \ldots, X_{1n_1} \sim F(\mu_1, \Sigma_1)$$
$$X_{21}, X_{22}, \ldots, X_{2n_2} \sim G(\mu_2, \Sigma_2)$$

Assumptions:

- The fourth moments of the coordinate variables in F and G are uniformly bounded.
- For $W = X_{1i} - X_{1j}$ or $X_{2i} - X_{1j}$ or $X_{1i} - X_{2j}$, with

$$W = \big(W^{(1)}, W^{(2)}, \ldots, W^{(d)}\big),$$
$$\sum_r \sum_s \big|\mathrm{Corr}\big(W^{(r)}, W^{(s)}\big)\big| = o(d^2).$$

- There exist constants μ^2, σ_1^2 and σ_2^2 such that

$$\frac{1}{d}\|\mu_1 - \mu_2\|^2 \to \mu^2 \quad \text{as } d \to \infty$$
$$\frac{1}{d}\,\mathrm{tr}(\Sigma_1) \to \sigma_1^2 \quad \text{as } d \to \infty$$
$$\frac{1}{d}\,\mathrm{tr}(\Sigma_2) \to \sigma_2^2 \quad \text{as } d \to \infty$$

Under the first two assumptions, we have WLLN for $\{(W^{(h)})^2 : h \geq 1\}$:

$$\left|\frac{1}{d}\sum (W^{(h)})^2 - \frac{1}{d}\sum E(W^{(h)})^2\right| \xrightarrow{P} 0 \quad \text{as } d \to \infty$$
$$\left|\frac{1}{d}\|W\|^2 - E\frac{1}{d}\|W\|^2\right| \xrightarrow{P} 0 \quad \text{as } d \to \infty$$

This holds for $W = X_{1i} - X_{1j}$, $X_{2i} - X_{2j}$ or $X_{1i} - X_{2j}$. So, we have convergence of the scaled version of the squared Euclidean distance as before, to its mean, so one can show:

Under the third assumption, one can show that

$$\frac{1}{d}\,\mathrm{tr}(\Sigma_2) \to \sigma_2^2 \quad \text{as } d \to \infty$$

$$\frac{1}{d}E\|W\|^2 \longrightarrow \begin{cases} 2\sigma_1^2 & \text{if } W = X_{1i} - X_{1j} \text{ for some } i \neq j \\ 2\sigma_2^2 & \text{if } W = X_{2i} - X_{2j} \text{ for some } i \neq j \\ \sigma_1^2 + \sigma_2^2 + \mu^2 & \text{if } W = X_{1i} - X_{2j} \text{ for some } i, j \end{cases}$$

3 Fisher's Approach for LDA

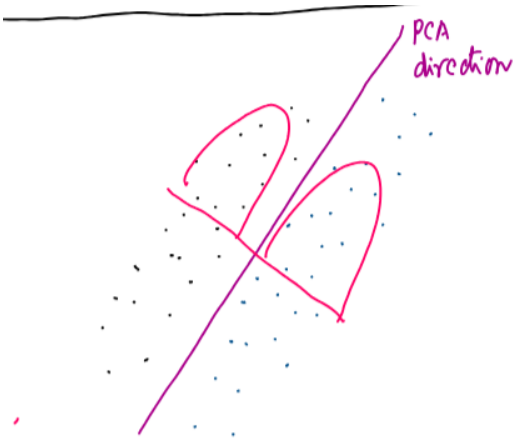
Let

$$X_1, X_2, \ldots, X_{n_1} : \text{ obs. from first population}$$
$$Y_1, Y_2, \ldots, Y_{n_2} : \text{ obs. from second population}$$

If we project the observations along a direction α, we get projected obs. from two populations

$$\left.\alpha' X_1, \alpha' X_2, \ldots, \alpha' X_{n_1}\right\}$$

$$\left.\alpha' Y_1, \alpha' Y_2, \ldots, \alpha' Y_{n_2}\right\}$$



One can think of finding α by maximizing

$$\frac{\text{Between group SS}}{\text{Within group SS}}$$

where

$$\text{Between group SS} \propto |\alpha' \bar{X} - \alpha' \bar{Y}|^2$$

$$\text{Within group} \propto \frac{\sum_{i=1}^{n_1} (\alpha' X_i - \alpha' \bar{X})^2 + \sum_{i=1}^{n_2} (\alpha' Y_i - \alpha' \bar{Y})^2}{n_1 + n_2}$$

Therefore,

$$\frac{\text{Between group SS}}{\text{Within group SS}} = \frac{\alpha'(\bar{X} - \bar{Y})(\bar{X} - \bar{Y})'\alpha}{\left[\alpha' \sum_{i=1}^{n_1} (X_i - \bar{X})(X_i - \bar{X})'\alpha + \alpha' \sum_{i=1}^{n_2} (Y_i - \bar{Y})(Y_i - \bar{Y})'\alpha\right] / (n_1 + n_2)}$$

$$= \frac{\alpha'(\bar{X} - \bar{Y})(\bar{X} - \bar{Y})'\alpha}{\alpha' S \alpha}$$

where

$$S = \frac{1}{n_1 + n_2} \left[\sum_{i=1}^{n_1} (X_i - \bar{X})(X_i - \bar{X})' + \sum_{i=1}^{n_2} (Y_i - \bar{Y})(Y_i - \bar{Y})'\right]$$

Maximization of this quantity w.r.t. α gives the maximizer

$$\alpha_{\max} \propto S^{-1}(\bar{X} - \bar{Y})$$

Max value = squared Mahalanobis distance between $\bar{X}$ & $\bar{Y}$.

Derivation via Cauchy–Schwarz: Let $u = S^{1/2}\alpha$, $v = S^{1/2}(\bar{X} - \bar{Y})$.

$$(u'v)^2 \leq (u'u)(v'v)$$

$$\left(\alpha'(\bar{X} - \bar{Y})\right)^2 \leq \alpha' S \alpha \cdot (\bar{X} - \bar{Y})' S^{-1} (\bar{X} - \bar{Y})$$

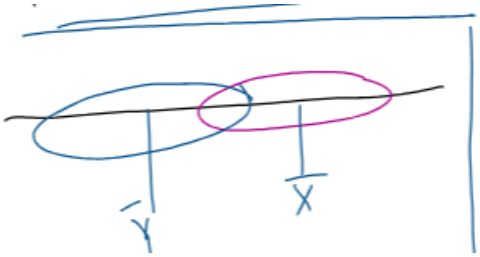
$$u \propto v \;\Rightarrow\; S^{1/2}\alpha \propto S^{1/2}(\bar{X}-\bar{Y}) \;\Rightarrow\; \alpha \propto S^{-1}(\bar{X}-\bar{Y})$$

$$\alpha' X_i\text{'s}, \quad \alpha' Y_j\text{'s} \qquad \Longrightarrow \qquad \alpha \propto S^{-1}(\bar{X}-\bar{Y})$$

4 Fisher's Direction and the LDA Decision Rule

Projected along a direction $\alpha = c\,S^{-1}(\bar{X}-\bar{Y})$ for some $c > 0$:

$$\alpha' X_1, \ldots, \alpha' X_{n_1} \qquad \alpha' Y_1, \ldots, \alpha' Y_{n_2}$$



Since $c > 0$:

$$\alpha'(\bar{X}-\bar{Y}) > 0 \quad \text{if } c > 0$$
$$\alpha'\bar{X} - \alpha'\bar{Y} = \alpha'(\bar{X}-\bar{Y}) = c(\bar{X}-\bar{Y})' S^{-1}(\bar{X}-\bar{Y}) > 0$$

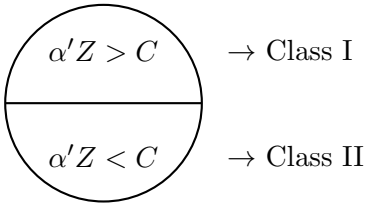


Figure 1: Fisher's classification rule based on the projected value.

How to find c?

$$Z' S^{-1}(\bar{X}-\bar{Y}) > C_0$$

Fisher's approach:

$$\alpha' Z > C \;\Rightarrow\; \text{Class I}, \qquad \alpha' Z < C \;\Rightarrow\; \text{Class II}, \qquad \alpha \propto S^{-1}(\bar{X}-\bar{Y})$$

LDA rule:

$$\left(Z - \frac{\bar{X}+\bar{Y}}{2}\right)' S^{-1}(\bar{X}-\bar{Y}) > \log\left(\frac{\pi_2}{\pi_1}\right)$$

Fisher's approach gives the **same direction** for the discriminating surface as what is used in LDA.

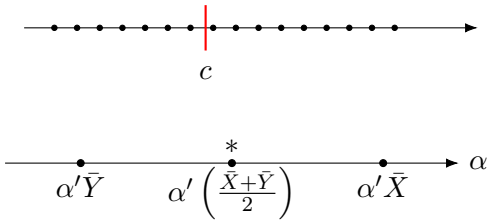


Figure 2: Threshold c on the projected line (top); midpoint interpretation matching LDA (bottom). This matches with LDA for $\pi_1 = \pi_2 = 1/2$.

$$\pi_1 \frac{1}{n_1} \sum_{i=1}^{n_1} \mathbb{I}\{X_{1i} < c\} \; + \; \pi_2 \frac{1}{n_2} \sum_{i=1}^{n_2} \mathbb{I}\{X_{2i} > c\}$$

$$\alpha' X_1, \ldots, \alpha' X_{n_1} \qquad \alpha' Y_1, \ldots, \alpha' Y_{n_2}$$

$$\frac{1}{n_1 n_2} \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} \mathbb{I}\{\alpha' X_i > \alpha' Y_j\} \qquad \sup_c \left| \frac{1}{n_1} \sum \mathbb{I}\{\alpha' X_i < c\} - \frac{1}{n_2} \sum \mathbb{I}\{\alpha' Y_i < c\} \right|$$

related to Half space depth *related to regression depth*

$$\frac{|\alpha' \bar{X} - \alpha' \bar{Y}|}{\sqrt{\alpha' S \alpha}}$$

$$\mu_1 = \mu_2 \implies \alpha' \mu_1 = \alpha' \mu_2 \;\; \forall \alpha \implies \alpha'(\mu_1 - \mu_2) = 0 \implies \mu_1 - \mu_2$$

5 Regression Approach to Classification

$$X_1, X_2, \ldots, X_d \quad \text{(features)}, \qquad Y \text{ (class label)}$$

X_1	X_2	$\ldots$	X_d	Class / Y
X_{11}	X_{21}	$\ldots$	X_{d1}	A $\to 1$
X_{12}	X_{22}	$\ldots$	X_{d2}	B $\to 0$
$\vdots$	$\vdots$		$\vdots$	$\vdots$
X_{1n}	X_{2n}	$\ldots$	X_{dn}	A $\to 1$

$$\sum_{i=1}^{n} \big(Y_i - f(X_{1i}, \ldots, X_{di})\big)^2$$

e.g. $f(X_{1i} \ldots X_{di}) = \beta_0 + \sum_{j=1}^{d} \beta_j X_{ji}$

We can estimate f from the data $\Big(\hat{f}(X_{i1} \ldots X_{id}) = \hat{\beta}_0 + \sum \hat{\beta}_j X_{ji}$ in this case of linear regression$\Big)$.

Classification rule: Assign an observation $X = (X_1 \ldots X_d)$ to class A if $\hat{f}(X) > 0.5$, and to class B, otherwise.

When we do least square regression, we estimate the conditional expectation of Y given $X_1 \ldots X_d$.

Here

$$E(Y \mid X_1 \ldots X_d) = 1.\, P(Y = 1 \mid X_1 \ldots X_d) + 0.\, P(Y = 0 \mid X_1 \ldots X_d) = P(A \mid X_1 \ldots X_d)$$

$E\big(Y - f(X_1 \ldots X_d)\big)^2$ is minimized when $f(X_1 \ldots X_d) = E(Y \mid X_1 \ldots X_d)$

$$\hat{f}(X_1 \ldots X_d) = \hat{P}(A \mid X_1 \ldots X_d)$$

So, if $\hat{f}(X_1 \ldots X_d) > 0.5$, it is reasonable to classify $X = (X_1 \ldots X_d)$ to class A.

But if f is linear, this probability estimate may take values outside the $[0, 1]$ interval.

Kernel Regression

$$\hat{f}(X_1 \ldots X_d) = \sum_{i=1}^{n} K\left(\frac{X - X_i}{h}\right) Y_i \Big/ \sum_{i=1}^{n} K\left(\frac{X - X_i}{h}\right)$$

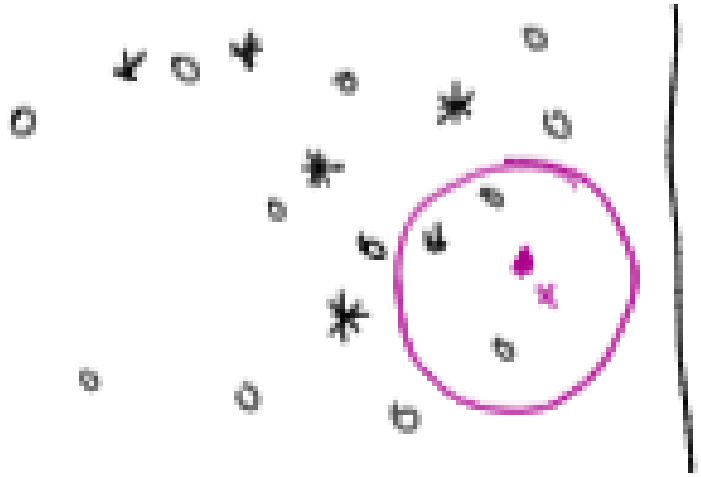
NN Regression:

$$= \frac{\frac{1}{n} \sum\limits_{i:Y_i=1} K\left(\frac{X - X_i}{h}\right)}{\frac{1}{n} \sum\limits_{i:Y_i=0} K\left(\frac{X - X_i}{h}\right) + \frac{1}{n} \sum\limits_{i:Y_i=1} K\left(\frac{X - X_i}{h}\right)}$$

Regression tree is another such method.

NN Regression

$$\hat{f}_k(X) = \frac{\sum Y_i}{k}, \qquad X_i \text{ in set of } k \text{ neighbours of } X$$



$$= \frac{\frac{1}{h^d}\left(\frac{n_A}{n} \cdot \frac{1}{n_A} \sum_{i:Y_i=1} K\left(\frac{X - X_i}{h}\right)\right)}{\frac{1}{h^d}\left[\frac{n_B}{n} \cdot \frac{1}{n_B} \sum_{i:Y_i=0} K\left(\frac{X - X_i}{h}\right) + \frac{n_A}{n} \cdot \frac{1}{n_A} \sum_{i:Y_i=1} K\left(\frac{X - X_i}{h}\right)\right]}$$

$$= \frac{\hat{\pi}_A \hat{f}_{Ah}(X)}{\hat{\pi}_B \hat{f}_{Bh}(X) + \hat{\pi}_A \hat{f}_{Ah}(X)}$$

3-NN estimate of f at $X = \dfrac{\text{No. of obs. from A in the set of 3 neighbours}}{3}$
So, kernel regression gives the same rule as KDA if the priors are estimated by the sample proportions.

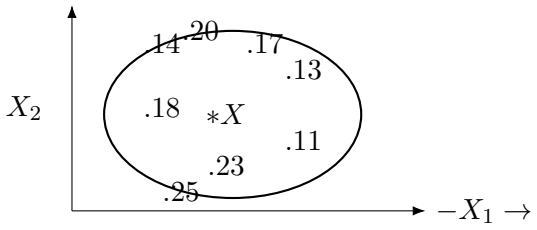


Figure 3: 6-NN estimate construction around point X.

6-NN estimate of f at X is

$$\hat{f}(X) = \frac{5 + 20 + 17 + 13 + 23 + 11}{6}$$

6 From Linear to Nonlinear Regression Forms

$$\begin{aligned}\text{Lin: } & f(X_1 \dots X_d) = \beta_0 + \beta_1 X_1 + \cdots + \beta_d X_d \\ \text{Additive: } & f(X \dots X_d) = f_1(X_1) + \cdots + f_d(X_d) \\ \text{Projection Pursuit: } & f(X \dots X_d) = f_1(\alpha_1' X) + f_2(\alpha_2' X) + \cdots + f_k(\alpha_k' X)\end{aligned}$$

We cannot ensure that the estimates will be in $[0, 1]$ for all values of X.

$$f(X_1 \dots X_d) = \sigma(\beta_0 + \beta_1 X_1 + \cdots + \beta_d X_d) \approx \text{GLM} = \text{SLP}, \quad \text{where } \sigma(t) = \frac{e^t}{1+e^t}$$

$$f(X \dots X_d) = \sigma\big(f_1(X_1) + \cdots + f_d(X_d)\big) \approx \text{GAM}$$

$$f(X \dots X_d) = \sigma\big(f_1(\alpha_1' X) + \cdots + f_K(\alpha_K' X)\big) \approx \text{MLP with one hidden layer}$$

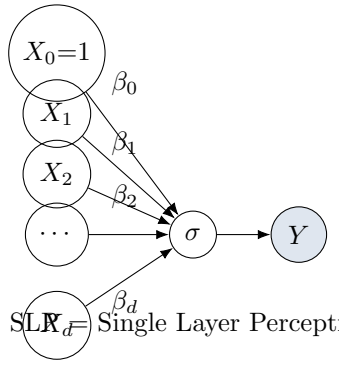
Let us consider the case

$$f(X \dots X_d) = \sigma(\beta_0 + \beta_1 X_1 + \cdots + \beta_d X_d)$$

$$\text{Min} \sum_{i=1}^{n} \big(Y_i - f(X_{1i} \dots X_{id})\big)^2 = \text{Min} \sum \left(Y_i - \frac{e^{\beta_0 + \beta_1 X_{1i} + \cdots + \beta_d X_{di}}}{1 + e^{\beta_0 + \beta_1 X_{1i} + \cdots + \beta_d X_{di}}}\right)^2 \quad \leftarrow \text{Can we do this?}$$

$$\hat{f}(X \dots X_d) = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 X_1 + \cdots + \hat{\beta}_d X_d}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 X_1 + \cdots + \hat{\beta}_d X_d}} > 0.5 \ ?$$

This can be viewed as SLP with squared error loss.



SLP = Single Layer Perceptron

Figure 4: Single Layer Perceptron (SLP) representation of logistic regression.

6.1 Newton–Raphson Method

$f'(X^*) = 0$: Assume

$$\underbrace{f'(X^*)}_{0} \approx f'(X_0) + (X_* - X_0)\, f''(X_0)$$

$$(X_* - X_0) = -\frac{f'(X_0)}{f''(X_0)}$$

$$\Rightarrow X_* = X_0 - \frac{f'(X_0)}{f''(X_0)}$$

$$X_1 = X_0 - H^{-1}\nabla f(X_0)$$

6.2 Minimization: Two Options

$$\min f(X)$$

Option 1: $f'(x) = 0 \leftarrow$ solve X

Option 2 (Gradient Descent):

$$X_0, \qquad X_1 = X_0 - \lambda \nabla f(X_0)$$

Usually, the gradient descent method is used in SLP.

7 Logistic Regression as a Likelihood-Based Model

Response	Output
Y_1	$\sigma(\beta_0 + \beta_1 X_{11} + \cdots + \beta_d X_{d1})$
$\vdots$	$\vdots$
Y_n	$\sigma(\beta_0 + \beta_1 X_{1n} + \cdots + \beta_d X_{dn})$

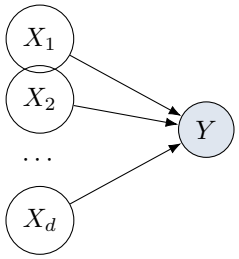


Figure 5: Y as a function of predictors $X_1, \ldots, X_d$.

$$P(Y = 1) = \sigma(\beta_0 + \beta_1 X_1 + \cdots + \beta_d X_d)$$

assumption:

$$Y \mid X_1 \ldots X_d \sim \text{Ber}\big(\sigma(\beta_0 + \cdots + \beta_d X_d)\big)$$

Instead of squared error loss, we can consider other loss functions. **"Softmax"** is one of them.

Likelihood Derivation

$$\text{Likelihood} = \prod_{i=1}^{n} f(Y_i, X_{1i} \ldots X_{di}) = \prod_{i=1}^{n} f_X(X_{1i} \ldots X_{di})\, f_{Y|X}(Y_i)$$

$$= \underbrace{\prod_{i=1}^{n} f_X(X_{1i} \dots X_{di})}_{\substack{\text{Dist. of } (X_1 \dots X_d) \\ \text{does not involve } \beta_0, \beta_1, \dots, \beta_d}} \prod_{i=1}^{n} f_{Y|X}(Y_i)$$

$$= \prod_{i=1}^{n} \Big[\sigma(\beta_0 + \dots + \beta_d X_{di})\Big]^{Y_i} \Big[1 - \sigma(\beta_0 + \dots + \beta_d X_{di})\Big]^{1-Y_i}$$

Check whether minimization of the softmax loss is same as maximization of the likelihood function.

8 Multi-Class Classification

What happens if there are more than two classes?

$X_1 \dots X_d$	Class labels	Y_1	Y_2	Y_3
...	A	1	0	0
...	B	0	1	0
...	C	0	0	1
...	B	0	1	0
...	C	0	0	1
...	A	1	0	0

$$\frac{\sum K\left(\frac{X - X_i}{h}\right) Y_{1i}}{\sum K\left(\frac{X - X_i}{h}\right)} = \hat{P}(A \mid X)$$

$$\frac{\sum K\left(\frac{X - X_i}{h}\right) Y_{2i}}{\sum K\left(\frac{X - X_i}{h}\right)} = \hat{P}(B \mid X)$$

$$\frac{\sum K\left(\frac{X - X_i}{h}\right) Y_{3i}}{\sum K\left(\frac{X - X_i}{h}\right)} = \hat{P}(C \mid X)$$

Classify X to the class having the largest estimated posterior.
If $\hat{P}(A \mid X) > \hat{P}(B \mid X)$ & $1 - \hat{P}(A \mid X) - \hat{P}(B \mid X)$, we classify X to class A.

$$X_1 X_2 \dots X_d \quad Y_1\ Y_2\ Y_3$$

$$E(Y_1 \mid \cdots) + E(Y_2 \mid \cdots) + E(Y_3 \mid \cdots) = 1$$

$$X_0\, X_1 \dots X_d \qquad \text{(HW)}$$